

Photoionization Cross Sections in EXHALE:

H I, He I, He II, and the He I 2³S Triplet,

with Benchmarks against Verner+1996 and Norcross 1971

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1. Overview

EXHALE computes the photoionization rate of each species by integrating its energy-dependent cross section $\sigma(E)$ against the local stellar spectrum on a fixed energy grid (the cross sections are pre-tabulated in `set_energy_vectors.f90`). All cross-section routines live in `src/modules/functions/cross_sec.f90` (module `Cross_sections`) and return σ in units of $1 \text{ Mb} \equiv 10^{-18} \text{ cm}^2$, with photon energy E in eV.

This memo (i) collects the four atomic cross sections relevant to the He 10830 / H α / Ly α diagnostics — H I, He I, He II, and the metastable He I 2³S triplet — and (ii) benchmarks the two *fitted* helium cross sections against independent references:

- **He I ground state:** the EXHALE two-term ATES fit (`sigma_HeI`) versus the Verner, Ferland, Korista & Yakovlev (1996; hereafter VFKY96) single-shell analytic fit, over 24.6–500 eV.

- **He I 2³S**: the EXHALE broken power-law fit (sigma_HeI23S) versus the tabulation of Norcross (1971) data shipped with the p-winds code, over 4.77–60 eV.

All numbers and figures below are produced by the companion script `docs/compare_photoion_cross_sections.py` (§7.), which ports the Fortran routines to Python verbatim and writes vector-PDF figures into `docs/figures/`.

2. The four cross sections

2.1 Hydrogenic atoms: H I and He II

H I ($Z = 1$) and He II ($Z = 2$) use the analytic hydrogenic cross section ($\sigma(E, Z)$), with threshold $E_0 = 13.6Z^2$ eV. For $E > E_0$,

$$\sigma(E) = \frac{6.3}{Z^2} \left(\frac{E_0}{E} \right)^4 \frac{\exp[4 - 4 \arctan(\epsilon)/\epsilon]}{1 - \exp(-2\pi/\epsilon)} \text{ Mb}, \quad \epsilon = \sqrt{E/E_0 - 1}, \quad (1)$$

which is the exact non-relativistic result with the Gaunt-factor correction; at threshold $\sigma = 6.3/Z^2$ Mb (6.3 Mb for H I, 1.575 Mb for He II).

2.2 He I ground state (two-term fit)

The He I ground state uses the original ATES two-term fit (sigma_HeI), valid above $E_{\text{th}} = 24.6$ eV:

$$\sigma_{\text{HeI}}(E) = \frac{0.6935}{(E/100)^{1.82} + (E/100)^{3.23}} \text{ Mb}. \quad (2)$$

2.3 He I 2³S metastable (broken power law)

The metastable triplet — the lower level of the He 10830 line — is photoionized by photons with $\lambda \lesssim 2593$ Å ($E \gtrsim 4.78$ eV). EXHALE uses a four-segment broken power law in $\log \sigma$ – $\log E$ (sigma_HeI23S), fit to the data of Norcross (1971). The breakpoints sit at $\lambda = 2593.01, 1655.63, 357.34, 271.94, 209.49$ Å (i.e. $E = 4.78, 7.49, 34.70, 45.59, 59.18$ eV); the cross section is set to zero outside $[4.78, 59.18]$ eV. The third segment rises steeply to reproduce the deep Cooper-type minimum near 35 eV followed by the sharp rise toward ~ 45 eV.

2.4 Verner+1996 single-shell fit (VFKY96)

For benchmarking we use the VFKY96 analytic fit, which is the form EXHALE already adopts for metals (sigma_VFKY96); for $E \geq E_{\text{th}}$,

$$x = E/E_0 - y_0, \quad z = \sqrt{x^2 + y_1^2}, \quad Q = 5.5 - P/2, \quad (3)$$

$$\sigma(E) = \sigma_0 [(x-1)^2 + y_w^2] z^{-Q} (1 + \sqrt{z/y_a})^{-P} \text{ Mb}. \quad (4)$$

Verner+1996 Table 1 provides ground-shell parameters for *all* ions $Z = 1$ –30, including H I, He I and He II (Table 1); the He I 2³S excited level is *not* in that table, which is why a separate Norcross-based fit is required for the triplet.

Table 1: VF KY96 ground-shell parameters (Verner+1996 Table 1) used here. σ_0 in Mb.

ion	E_{th}	E_0	σ_0	y_a	P	y_w	(y_0, y_1)
H I	13.60	0.4298	5.475×10^4	32.88	2.963	0.0	(0, 0)
He I	24.59	13.61	949.2	1.469	3.188	2.039	(0.4434, 2.136)
He II	54.42	1.720	1.369×10^4	32.88	2.963	0.0	(0, 0)

3. Overview comparison

Figure 1 shows the four EXHALE-adopted cross sections on a common log–log scale. The ordering of thresholds (He 2³S at 4.78, H I at 13.6, He I at 24.6, He II at 54.4 eV) and the peak magnitudes (~ 5 –8 Mb near threshold for H I, He I and He 2³S; 1.6 Mb for He II) are evident. The He 2³S curve displays the characteristic minimum near 35 eV and the rise toward 45 eV inherited from the Norcross data.

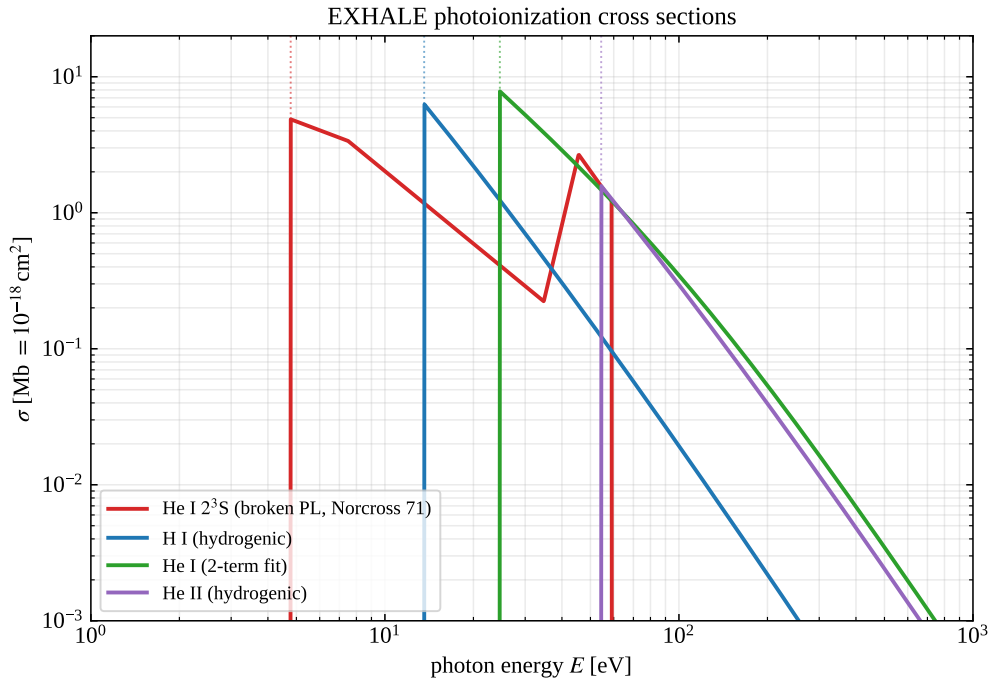


Figure 1: Photoionization cross sections adopted in EXHALE for H I, He I, He II and the metastable He I 2³S triplet. Dotted vertical lines mark the ionization thresholds.

4. He I ground state: EXHALE two-term fit vs Verner+1996

Figure 2 and Table 2 compare the EXHALE two-term fit with the VF KY96 He I cross section over 24.6–500 eV. The two agree to within $\sim 14\%$ everywhere: the fits match within a few percent at threshold and at high energy, with the two-term form running ~ 12 – 14% below VF KY96 in the 50–100 eV range (the band that dominates He photoionization in a typical XUV field). The median ratio over the interval is 0.94.

Table 2: He I cross section (Mb): EXHALE two-term fit vs Verner+1996.

E [eV]	EXHALE	Verner	ratio
24.6	7.821	7.430	1.053
30	5.244	5.361	0.978
50	1.779	2.022	0.880
100	0.347	0.394	0.880
200	0.054	0.055	0.973
300	0.0165	0.0160	1.028
500	0.0035	0.0032	1.089
ratio over 24.6–500 eV: min 0.861, max 1.089, median 0.939			

Because both are accurate fits to the same underlying He I data, the $\leq 14\%$ difference is well within the photoionization-rate uncertainty budget. Replacing `sigma_HeI` with the VFKY96 He I parameters of Table 1 (a one-line wrapper around the existing `sigma_VFKY96`) would unify He I with the metal and He II treatment and slightly improve the 50–100 eV band, but is not required for the ionization balance.

5. He I 2³S: EXHALE broken power law vs p-winds/Norcross

The p-winds code (dos Santos et al. 2022) ships the He 2³S photoionization cross section as a 23-point table of Norcross (1971), converting a differential oscillator strength $df/d\epsilon$ to a cross section via $\sigma[\text{cm}^2] = 8.0670 \times 10^{-18} (df/d\epsilon)$. Crucially, the EXHALE broken-power-law breakpoints coincide *exactly* with the Norcross wavelength nodes, confirming that `sigma_HeI23S` is a fit to this same data set.

Figure 3 overlays the EXHALE curve, the tabulated Norcross points, and their log–log interpolation; Table 3 lists the comparison. The agreement is excellent: the broken power law reproduces the table to within $\sim 10\%$ across 4.78–59 eV, with a median ratio of 1.00. The only localized deviation ($\sim 15\%$) occurs right at the Cooper minimum near 30 eV, where the cross section is smallest and therefore least important for the rate.

5.1 Cross-check against TOPbase / Opacity Project

As an independent, fully *ab initio* reference we extracted the He I 2³S ($1s2s^3S$) photoionization cross section from the **TOPbase** (Opacity Project) R-matrix database at the CDS server. The relevant level is $NZ = 2$, $NE = 2$, $i\text{SLP} = 300$ (3S , even parity), $i\text{LV} = 1$; the table has 922 points with the photon energy in Rydberg and σ in Mb (level energy -0.35032 Ry, i.e. ionization threshold 4.766 eV). The query URL and parsing are in the companion script (§7.); the raw table is saved as `docs/data/topbase_HeI_2_3S_photo.dat`.

Two points must be kept in mind when comparing. (i) The TOPbase threshold sits at the *theoretical* energy (4.63 eV for the first point), ~ 0.14 eV below the experimental 4.77 eV; this is the usual OP threshold offset. (ii) Above ~ 35 eV TOPbase resolves a dense forest of **autoionizing resonances** (doubly-excited states converging on the excited He II

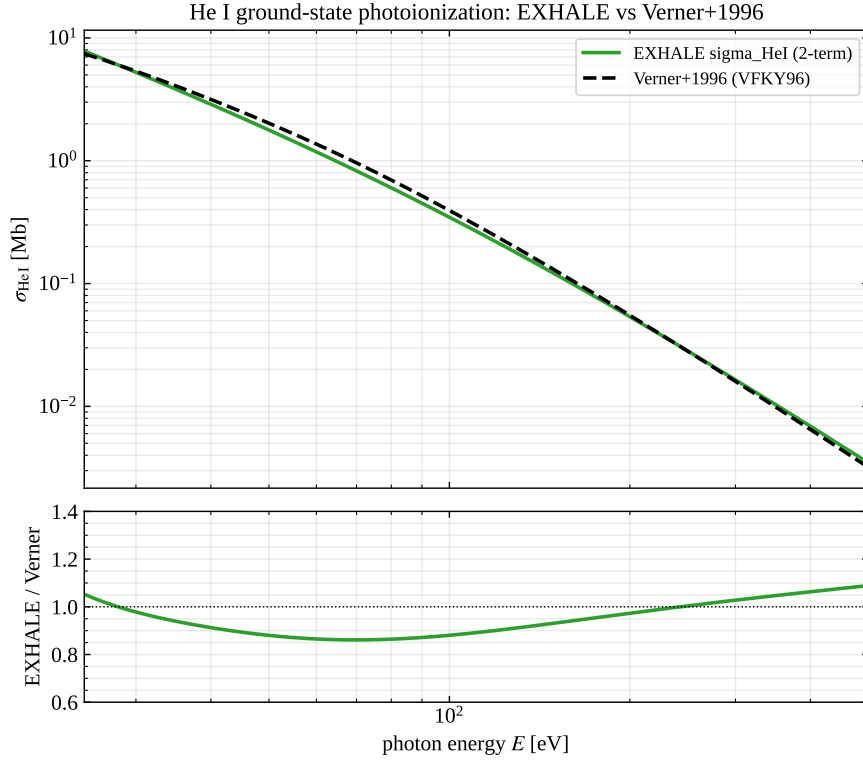


Figure 2: He I ground-state photoionization cross section: EXHALE two-term fit (sigma_HeI, green) vs the Verner+1996 single-shell fit (black dashed). Bottom panel: ratio EXHALE/Verner.

core), with peaks reaching $\sigma \simeq 3200$ Mb. These are real but extremely narrow, and the smooth Norcross/EXHALE fits represent only their resonance-averaged background.

Figure 4(a) shows the full range with the resonance structure; panel (b) zooms into the smooth near-threshold region (4.6–25 eV). In that smooth region the three independent descriptions agree to within $\sim 8\%$ (Table 4): EXHALE/TOPbase has min 0.92, max 1.07, median 1.00 over 5–20 eV. This confirms that the EXHALE broken power law — and the Norcross data it is fit to — reproduces the OP R-matrix background where it matters for the He 2³S population.

Two caveats remain for completeness: (a) EXHALE sets $\sigma_{\text{He } 2^3\text{S}} = 0$ above 59.2 eV, whereas TOPbase shows a smooth $\sim E^{-3}$ tail continuing to several hundred eV (small in magnitude, $\lesssim 0.1$ Mb); (b) the broken-power-law “bump” near 45 eV is a smooth stand-in for the resonance complex and should not be read as a true cross section at any single energy there.

5.2 Does the > 59 eV truncation matter for the rate?

The relevant quantity for the He 2³S population (and hence He 10830) is the *photoionization rate*, $\Gamma_{2^3\text{S}} \propto \int \sigma_{2^3\text{S}}(E) N(E) dE$, where $N(E) = J_\nu / h\nu \propto J_{\text{inc}}(E)/E$ is the photon-number flux. In the EXHALE power-law SED, $J_{\text{inc}} \propto E^{\text{PLind}}$ over the EUV band $[e_{\text{low}}, e_{\text{mid}}] = [4.8, 124]$ eV (normalized to L_{EUV}) and the X-ray band $[124, 1240]$ eV (to L_X), so $N(E) \propto E^{\text{PLind}-1}$. Because both $N(E)$ and $\sigma_{2^3\text{S}}$ fall steeply, the integrand peaks at the threshold; the rate is dominated by the near-threshold UV (~ 4.8 –15 eV), where σ is largest (1–5 Mb) and photons are most numerous.

Table 3: He I 2³S cross section (Mb): EXHALE broken PL vs p-winds/Norcross. (At threshold 4.78 eV the table interpolation is undefined; EXHALE gives 4.867 Mb, matching the Norcross threshold value $8.0670 \times 0.605 = 4.881$ Mb.)

E [eV]	EXHALE	p-winds	ratio
6	4.046	4.092	0.989
8	3.006	3.110	0.966
12	1.465	1.463	1.002
20	0.593	0.546	1.085
30	0.289	0.277	1.042
45	2.382	2.442	0.975
55	1.517	1.516	1.001
59	1.226	1.251	0.980
ratio over 4.78–59 eV: min 0.853, max 1.095, median 1.000			

Table 4: He I 2³S cross section (Mb) in the smooth near-threshold region: EXHALE vs Norcross/p-winds vs TOPbase.

E [eV]	EXHALE	p-winds	TOPbase	EX/TB	PW/TB
5.0	4.693	4.671	5.081	0.924	0.919
6.0	4.046	4.092	4.231	0.956	0.967
8.0	3.006	3.110	2.866	1.049	1.085
10.0	2.024	2.069	2.025	1.000	1.022
12.0	1.465	1.463	1.488	0.985	0.983
16.0	0.880	0.872	0.877	1.004	0.994
20.0	0.593	0.546	0.574	1.033	0.952
EXHALE/TOPbase over 5–20 eV: min 0.924, max 1.066, median 0.999					

Table 5 gives the relative increase of Γ_{2^3S} when σ_{2^3S} is extended above 59.18 eV with the TOPbase tail (integrated to 323 eV). The effect is only 0.1–1.2% across realistic spectral indices (PLind = -1 to -2) and X-ray/EUV ratios; for the EUV slopes typical of cool stars (PLind $\simeq -1.3$ to -1.6) it is 0.3–0.6%. This is far below the $\sim 10\%$ uncertainty of the cross-section fit itself (§5.) and of the composition/SED. The 35–59 eV resonance “bump” likewise carries only $\sim 2\%$ of the current rate.

Moreover, these are *upper bounds*: the power-law SED extrapolates the EUV slope down to 4.8 eV, whereas a real star has a much larger photospheric near-UV flux (4.8–13.6 eV); with a realistic SED the near-threshold region dominates even more and the > 59 eV contribution shrinks further. (Heating is also unaffected: the 2³S number density is negligible compared with the H/He ground states, so its photoionization deposits negligible energy regardless of the high-energy tail.) We therefore retain the 59 eV cutoff; extending it is a cosmetic, sub-percent refinement.

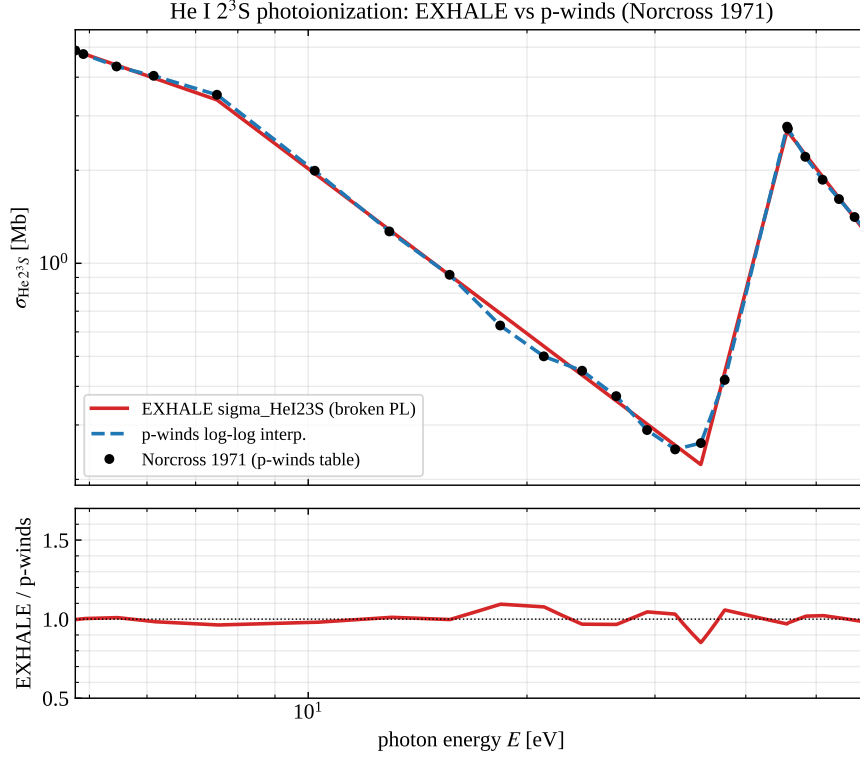


Figure 3: He I 2³S photoionization cross section: EXHALE broken power law (sigma_HeI23S, red) vs the p-winds log–log interpolation (blue dashed) of the Norcross (1971) table (black points). Bottom panel: ratio EXHALE/p-winds.

5.3 Impact of the autoionizing resonances

Between ~ 40 and 55 eV, TOPbase shows a dense forest of autoionizing resonances (doubly-excited states converging on the He II $n = 2$ threshold at 65.4 eV), with peaks reaching $\sigma \simeq 3223$ Mb, whereas the EXHALE broken power law represents this region by a smooth bump peaking at only ~ 2.4 Mb (Fig. 4a). It is tempting to ask whether the resonances, being so tall, substantially raise the He 2³S photoionization rate.

They do not — but the analysis requires care, because *the raw TOPbase table must not be integrated directly*. Integrating both cross sections on the *same* native TOPbase energy grid (so the quadrature is identical) gives the total-rate ratio TOPbase/EXHALE in Table 6. With no clipping the resonances appear to raise the rate by $\sim 13\%$ (at PLind = -1.3); but this increase is produced almost entirely by the 2–4 tallest grid points. The narrowest resonances ($\Delta E \sim 0.01$ eV) are sampled by only one or two points, so the trapezoidal rule assigns them a triangular area fixed by the arbitrary mesh spacing rather than by the physical Lorentzian width. Clipping σ at $\gtrsim 100$ Mb removes just these few unresolved spikes and the ratio collapses to 0.99 – 1.02 : **the robust net effect of the resonances on the rate is only ~ 1 – 2%** .

Physically this is expected: although individual resonances are very tall, they are extremely narrow and therefore intercept little photon flux. The resonance-*averaged* cross section $\langle \sigma \rangle = \sigma_{\text{bg}} + \sum_i (\text{resonance area})_i / \Delta E$ sits only modestly above the smooth background, which the few-Mb Norcross/EXHALE values already approximate. The net resonance effect (~ 1 – 2%) is comparable to the > 59 eV tail (§5.2) and well below the $\sim 10\%$ fit and composition/SED uncertainties, so it does not change the He 10830 conclusions.

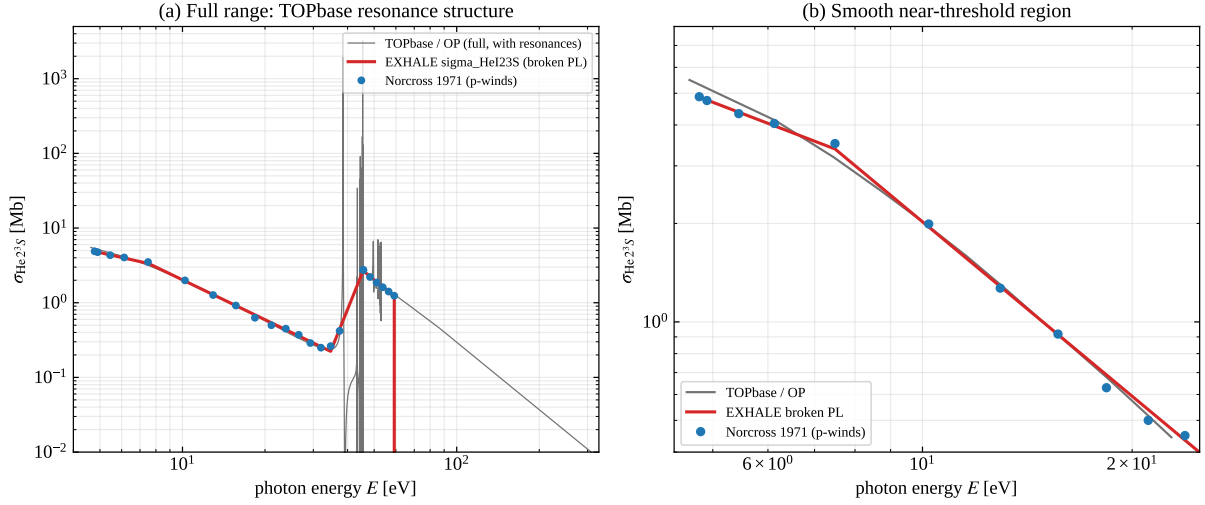


Figure 4: He I 2³S photoionization: EXHALE broken power law (red), Norcross/p-winds table (blue points), and TOPbase/OP (gray). **(a)** Full range: TOPbase resolves autoionizing resonances above ~ 35 eV (peaks to ~ 3200 Mb) that the smooth fits average over; EXHALE is truncated to zero above 59 eV. **(b)** Smooth near-threshold region: the three agree to within $\sim 8\%$.

Table 5: Relative increase (%) of the He 2³S photoionization rate when σ_{2^3S} is extended above 59 eV with the TOPbase tail, for an EXHALE power-law SED of index PLind and X-ray/EUV luminosity ratio L_X/L_{EUV} .

PLind	$L_X/L_{EUV} = 0.1$	0.3	1.0
-1.0	1.13%	1.15%	1.21%
-1.3	0.57%	0.59%	0.65%
-1.6	0.29%	0.30%	0.36%
-2.0	0.12%	0.13%	0.19%

Practical caveat: for percent-level work one must use a properly resonance-averaged Opacity-Project cross section (e.g. the resonance-averaged tabulations in the spirit of Bautista et al.); the raw 922-point TOPbase table should never be fed into a broadband rate integral, as the quadrature artifact in Table 6 illustrates.

6. Summary

- EXHALE uses three distinct prescriptions: an exact hydrogenic form for H I and He II, a two-term ATES fit for He I, and a Norcross-1971 broken power law for the He I 2³S triplet (the excited level absent from the Verner+1996 ground-shell table).
- The He I two-term fit agrees with Verner+1996 to within 14% (24.6–500 eV; median 0.94), running slightly low at 50–100 eV.
- The He I 2³S broken power law reproduces its parent Norcross/p-winds table to within $\sim 10\%$ (4.78–59 eV; me-

Table 6: Convergence/spike test of the resonance contribution. Total He 2³S rate ratio TOPbase/EXHALE (PLind = −1.3), both integrated on the same native grid, as the TOPbase cross section is clipped at an upper value; “pts>cap” counts the grid points removed.

σ clip [Mb]	TOPbase/EXHALE	pts > cap
none (raw)	1.134	0
500	1.024	2
200	1.001	2
100	0.993	4
50	0.989	7

dian 1.00), including the Cooper minimum and the rise near 45 eV.

- An independent check against the **TOPbase / Opacity Project** R-matrix cross section (§5.1) confirms agreement to $\sim 8\%$ in the smooth near-threshold region (5–20 eV; median 1.00). TOPbase additionally shows autoionizing resonances above ~ 35 eV (averaged over by the fits) and a smooth tail above 59 eV (where EXHALE truncates to zero).
- The 40–55 eV autoionizing resonances raise the He 2³S rate by only ~ 1 –2% once the few numerically under-resolved spikes are excluded (§5.3); the raw TOPbase table must not be integrated directly.
- Both fitted helium cross sections are adequate for the photoionization balance. Extending the He 2³S fit above 59 eV with the TOPbase tail changes its photoionization rate by only 0.1–1.2% (§5.2), so the cutoff is harmless; the one worthwhile refinement, if any, is to migrate He I to the VFKY96 form (Table 1) for internal consistency.

7. Reproducing this comparison

All figures and numbers are generated by docs/compare_photoion_cross_sections.py:

```
cd ATES/EXHALE/docs
python3 compare_photoion_cross_sections.py
# -> figures/xsec_overview.pdf
#   figures/xsec_HeI_vs_Verner.pdf
#   figures/xsec_HeI23S_vs_pwinds.pdf
#   figures/xsec_HeI23S_topbase.pdf
# reads: data/topbase_HeI_2_3S_photo.dat
```

The TOPbase table was retrieved (once) from the CDS Opacity Project server with

```
curl -L -G "https://cds.unistra.fr/cgi-bin/topbase/topbase.sh" \
  --data "com=dt" --data "ent=p" \
```

```
--data "nz1=2" --data "nz2=2" --data "ne1=2" --data "ne2=2" \  
--data "is1=3" --data "is2=3" --data "il1=0" --data "il2=0" \  
--data "ip1=0" --data "ip2=0" --data "lv1=1" --data "lv2=1" \  
--data "en1=0.0" --data "en2=0.0" --data "so=s1"
```

and parsed into data/topbase_HeI_2_3S_photo.dat (photon energy in Ry $\rightarrow$ eV; σ in Mb).

The script ports sigma, sigma_HeI, sigma_HeI23S and sigma_VFKY96 from cross_sec.f90 (using the same constants h_{ev} , c from parameters.f90), hard-codes the Verner+1996 He I/H I/He II parameters and the 23-point p-winds/Norcross table, and emits the three vector-PDF figures.